1. The natural atmospheric phenomenon that warms the Earth's surface by absorbing outgoing infrared radiation is the:

(A) Coriolis effect
(B) Greenhouse effect
(C) Doppler effect
(D) Tyndall effect

2. Which naturally occurring atmospheric gas contributes the largest percentage (approx 60%) to the natural greenhouse effect?

(A) Carbon dioxide (CO_2)
(B) Water vapor (H_2O)
(C) Methane (CH_4)
(D) Ozone (O_3)

3. The reference greenhouse gas against which the Global Warming Potential (GWP) of all other gases is standardized (GWP = 1) is:

(A) Methane (CH_4)
(B) Carbon dioxide (CO_2)
(C) Nitrous oxide (N_2O)
(D) Sulfur hexafluoride (SF_6)

4. Over a 100-year time horizon according to the IPCC Sixth Assessment Report (AR6), the GWP of Methane (CH_4) is approximately:

(A) 1
(B) 10
(C) 28
(D) 300

5. Which industrial gas used as an electrical insulator in high-voltage switchgear has an extreme 100-year GWP exceeding 23,500?

(A) Sulfur hexafluoride (SF_6)
(B) Carbon tetrachloride (CCl_4)
(C) Nitrogen dioxide (NO_2)
(D) Carbon monoxide (CO)

6. The stratospheric ozone layer absorbs and protects terrestrial life from harmful solar:

(A) Infrared radiation
(B) Shortwave ultraviolet (UV-B and UV-C) radiation
(C) Visible green light
(D) Radio waves

7. The thickness of stratospheric ozone is measured globally in terms of:

(A) Becquerels
(B) Dobson Units (DU)
(C) Decibels
(D) Torr

8. One Dobson Unit (DU) corresponds to a compressed pure ozone layer thickness at standard temperature and pressure (STP) of:

(A) 1.0 mm
(B) 0.1 mm
(C) 0.01 mm (10 micrometers)
(D) 0.001 mm

9. An 'Ozone Hole' over Antarctica is officially defined when total column ozone drops below:

(A) 500 DU
(B) 300 DU
(C) 220 DU
(D) 100 DU

10. The landmark international treaty signed in 1987 to phase out ozone-depleting chlorofluorocarbons (CFCs) is the:

(A) Kyoto Protocol
(B) Montreal Protocol
(C) Paris Agreement
(D) Basel Convention

11. The Kigali Amendment (2016) to the Montreal Protocol mandates the progressive global phase-down of which greenhouse gases?

(A) Hydrofluorocarbons (HFCs)
(B) Carbon dioxide (CO_2)
(C) Methane (CH_4)
(D) Nitrous oxide (N_2O)

12. The United Nations Framework Convention on Climate Change (UNFCCC) was adopted at which historic global summit?

(A) 1972 Stockholm Conference
(B) 1992 Rio Earth Summit
(C) 2002 Johannesburg Summit
(D) 2015 Paris Summit

13. The Kyoto Protocol (adopted in 1997) was historically significant because it established:

(A) Voluntary pledges for all developing nations
(B) Legally binding greenhouse gas emission reduction targets for Annex I industrialized nations
(C) A complete ban on fossil fuel mining
(D) A global carbon tax of per ton

14. The primary central goal of the 2015 Paris Agreement is to limit the increase in global average temperature to well below:

(A) 1.0 deg C
(B) 2.0 deg C above pre-industrial levels, striving for 1.5 deg C
(C) 3.5 deg C
(D) 5.0 deg C

15. Under the Paris Agreement, the non-binding post-2020 climate action plans submitted by individual countries are called:

(A) Clean Development Mechanisms
(B) Nationally Determined Contributions (NDCs)
(C) Kyoto Target Plans
(D) Carbon Border Adjustments

16. The Intergovernmental Panel on Climate Change (IPCC) was established in 1988 jointly by:

(A) WMO and UNEP
(B) WHO and UNESCO
(C) World Bank and WTO
(D) FAO and UNICEF

17. Which Article of the Constitution of India obligates the State to protect and improve the environment and safeguard forests?

(A) Article 21
(B) Article 48A (Directive Principles of State Policy)
(C) Article 14
(D) Article 370

18. Which Article of the Constitution of India imposes a Fundamental Duty on every citizen to protect and improve the natural environment?

(A) Article 51A(g)
(B) Article 19
(C) Article 32
(D) Article 300A

19. The Water (Prevention and Control of Pollution) Act was enacted by the Indian Parliament in which year?

(A) 1972
(B) 1974
(C) 1981
(D) 1986

20. The Air (Prevention and Control of Pollution) Act was enacted in India in:

(A) 1974
(B) 1981
(C) 1986
(D) 1991

21. The comprehensive umbrella legislation 'Environment (Protection) Act' was enacted in India in 1986 in the immediate aftermath of the:

(A) Chernobyl disaster
(B) Bhopal Gas Tragedy
(C) Love Canal crisis
(D) Minamata disaster

22. The Bhopal Gas Tragedy occurred on the night of December 2-3, 1984, due to the catastrophic release of which lethal gas?

(A) Phosgene
(B) Methyl Isocyanate (MIC)
(C) Hydrogen cyanide
(D) Chlorine

23. The Union Carbide factory in Bhopal was manufacturing which commercial carbamate pesticide using methyl isocyanate?

(A) DDT
(B) Sevin (Carbaryl)
(C) Malathion
(D) Glyphosate

24. The Forest (Conservation) Act was enacted in India in 1980 to ensure that no reserved forest land can be de-reserved or diverted without:

(A) Prior approval of the Central Government
(B) Approval from local village heads alone
(C) A permit from the State Police
(D) Permission from private logging corporations

25. The National Green Tribunal (NGT) was established under the NGT Act 2010 as a specialized judicial body headquartered in:

(A) Mumbai
(B) New Delhi
(C) Kolkata
(D) Chennai

26. Under the NGT Act, 2010, the Tribunal is mandated to dispose of environmental applications and appeals within what timeframe?

(A) 30 days
(B) 6 months from the date of filing
(C) 2 years
(D) 5 years

27. Which two foundational environmental law doctrines are explicitly mandated to be applied by the NGT in its rulings?

(A) Polluter Pays Principle and Precautionary Principle
(B) Absolute Sovereignty and Strict Laches
(C) Caveat Emptor and Laissez-Faire
(D) Eminent Domain and Police Power

28. Environmental Impact Assessment (EIA) in India is statutory governed under the EIA Notification 2006 issued under which Act?

(A) Wildlife Protection Act, 1972
(B) Environment (Protection) Act, 1986
(C) Water Act, 1974
(D) Factories Act, 1948

29. Under the EIA Notification 2006, industrial projects designated as 'Category A' require prior environmental clearance from the:

(A) State Environment Impact Assessment Authority (SEIAA)
(B) Central Ministry of Environment, Forest and Climate Change (MoEFCC)
(C) District Magistrate
(D) Central Pollution Control Board

30. The four sequential stages in the Prior Environmental Clearance process for Category B1 projects under EIA 2006 are:

(A) Screening, Scoping, Public Consultation, Appraisal
(B) Planning, Tendering, Construction, Demolition
(C) Extraction, Combustion, Refining, Inspection
(D) Survey, Notice, Fine, Closure

31. A tropical cyclone formed over the North Indian Ocean (Bay of Bengal and Arabian Sea) is fueled primarily by the latent heat of:

(A) Fossil fuel combustion
(B) Condensation of warm ocean water vapor (sea surface temperature > 26.5 deg C)
(C) Nuclear reactions in clouds
(D) Geothermal ocean venting

32. The spinning rotational deflection of tropical cyclones (counter-clockwise in the Northern Hemisphere) is governed by the:

(A) Magnus force
(B) Coriolis force
(C) Centrifugal force alone
(D) Gravitational pull of the moon

33. The calm, clear, low-pressure center of a mature tropical cyclone is termed the:

(A) Eye of the cyclone
(B) Eyewall
(C) Spiral rainband
(D) Storm surge apex

34. The devastating abnormal rise in sea water level accompanying a tropical cyclone landfall that causes coastal flooding is the:

(A) Tsunami wave
(B) Storm surge
(C) Tidal bore
(D) Seiche

35. The point on the Earth's surface directly vertically above the subterranean subterranean focus of an earthquake is the:

(A) Hypocenter
(B) Epicenter
(C) Fault scarp
(D) Seismic node

36. Which fast-moving seismic body waves propagate through solids, liquids, and gases via longitudinal compression?

(A) Primary (P) waves
(B) Secondary (S) waves
(C) Rayleigh waves
(D) Love waves

37. Shear seismic body waves (S-waves) can propagate ONLY through which physical state of matter?

(A) Solids only
(B) Liquids and gases only
(C) Gases only
(D) Plasma only

38. The logarithmic Richter Magnitude Scale measures an earthquake's:

(A) Intensity of surface damage observed by humans
(B) Total seismic wave energy released at the source
(C) Duration of fault shaking in minutes
(D) Depth of groundwater table change

39. In India, the National Disaster Management Authority (NDMA) is statutorily headed ex-officio by the:

(A) Union Minister of Home Affairs
(B) Prime Minister of India
(C) President of India
(D) Cabinet Secretary

40. The specialized multi-disciplinary armed force dedicated to disaster response and rescue in India under the DMA 2005 is the:

(A) CRPF
(B) NDRF (National Disaster Response Force)
(C) BSF
(D) ITBP

41. The simplified formula for Radiative Forcing (Delta F in W/m2) caused by increasing atmospheric CO2 concentration is:

(A) Delta F = 5.35 * ln(C / C_0)
(B) Delta F = 5.35 * (C - C_0)
(C) Delta F = C * C_0 / 5.35
(D) Delta F = 5.35 * exp(C / C_0)

42. In the radiative forcing equation, what does 'C_0' represent?

(A) The current atmospheric CO2 concentration
(B) The pre-industrial baseline CO2 concentration (approx 280 ppm)
(C) The total mass of oceanic carbon
(D) The solar constant

43. Doubling atmospheric CO2 concentration from 280 ppm to 560 ppm produces a direct radiative forcing of approximately:

(A) 1.2 W/m2
(B) 3.7 W/m2
(C) 7.4 W/m2
(D) 15.0 W/m2

44. In atmospheric ozone photochemistry, the classical four gas-phase reactions formulated in 1930 are known as the:

(A) Chapman mechanism
(B) Haber mechanism
(C) Arrhenius mechanism
(D) Fenton mechanism

45. A single chlorine free radical (Cl.) released from CFC photolysis can catalytically destroy approximately how many ozone molecules?

(A) 1 to 2 molecules
(B) 10 to 50 molecules
(C) 100,000 molecules before being sequestered in reservoir species
(D) Over 1 billion molecules

46. The two primary atmospheric chlorine 'reservoir species' that temporarily deactivate reactive chlorine radicals are:

(A) CF2Cl2 and CCl4
(B) Hydrochloric acid (HCl) and Chlorine nitrate (ClONO2)
(C) Ozone and water vapor
(D) Nitric oxide and oxygen

47. Polar Stratospheric Clouds (PSCs, Type I and II) accelerate Antarctic ozone hole formation during polar winter by:

(A) Providing cold ice crystal surfaces that catalyze the heterogeneous activation of reservoir chlorine into reactive Cl2
(B) Freezing all ozone molecules into solid ozone snow
(C) Blocking all sunlight permanently
(D) Absorbing all atmospheric nitrogen

48. Ocean Acidification occurs when anthropogenic CO2 dissolves in seawater, causing a decrease in pH and a sharp reduction in:

(A) Dissolved oxygen
(B) Carbonate ion ([CO3^2-]) concentration and calcium carbonate saturation state
(C) Chloride ion concentration
(D) Sodium concentration

49. The calcium carbonate saturation state Omega for aragonite in seawater is defined by which formula?

(A) Omega = [Ca2+] * [CO3^2-] / K'_sp
(B) Omega = [Ca2+] / [CO3^2-]
(C) Omega = K'_sp / ([Ca2+] * [CO3^2-])
(D) Omega = [H+] * [HCO3-]

50. Marine calcifying organisms (pteropods, coral reefs, shellfish) suffer skeletal shell dissolution when aragonite saturation Omega drops:

(A) Above 5.0
(B) Above 2.0
(C) Below 1.0 (under-saturation)
(D) Between 1.5 and 3.0

51. In the IPCC Sixth Assessment Report (AR6), future socioeconomic and technological pathways are modeled using:

(A) Shared Socioeconomic Pathways (SSPs) coupled with RCPs
(B) Malthusian population curves
(C) Pure free-market capital equations
(D) Planetary cycle loops

52. Under the Kyoto Protocol, the Clean Development Mechanism (CDM) allows Annex I industrialized nations to:

(A) Purchase Certified Emission Reduction (CER) credits by financing emission-reduction projects in developing nations
(B) Dump toxic wastes in international waters
(C) Build unmonitored coal power plants
(D) Cancel all financial debt

53. Article 6 of the Paris Agreement establishes international frameworks for cooperative climate approaches and market-based:

(A) International Transferred Mitigation Outcomes (ITMOs) and centralized crediting mechanisms
(B) Tariff-free fossil fuel transportation
(C) Compulsory military enforcement
(D) Bilateral mining concessions

54. Under the Water (Prevention and Control of Pollution) Act 1974, Section 33A grants State Pollution Control Boards the power to:

(A) Order the closure, prohibition, or regulation of any industrial operation and disconnect electricity/water supply
(B) Directly sentence corporate CEOs to death
(C) Impose trade tariffs on neighboring states
(D) Dissolve local municipal corporations

55. The Air (Prevention and Control of Pollution) Act was amended in 1987 to explicitly bring which pollutant within its regulatory scope?

(A) Radioactive radon
(B) Noise pollution
(C) Agricultural crop pollen
(D) Carbon dioxide

56. Under the Environment (Protection) Act, 1986, who is vested with statutory authority to take all necessary measures to protect the environment?

(A) The Central Government (Ministry of Environment, Forest and Climate Change)
(B) The United Nations Security Council
(C) Private industrial chambers of commerce
(D) State High Courts alone

57. Coastal Regulation Zone (CRZ) Notification divides coastal coastal stretches into distinct regulatory categories. CRZ-I encompasses:

(A) Highly developed urban coastal municipal ports
(B) Ecologically sensitive coastal areas (mangroves, coral reefs, turtle nesting sites) where new construction is strictly prohibited
(C) Rural coastal agrarian villages
(D) Open oceanic waters beyond 12 nautical miles

58. Under the Indian EIA Notification 2006, industrial projects classified as Category B1 must undergo:

(A) Full environmental impact assessment studies and mandatory public consultation
(B) Zero environmental review
(C) Only a verbal consultation with local police
(D) Immediate construction without permits

59. Under EIA 2006, which industrial activities are statutorily exempt from the mandatory Public Hearing stage?

(A) All Category A commercial petrochemical refineries
(B) Defense and national security projects, and offshore boundary projects
(C) All private real estate housing colonies
(D) Thermal power plants exceeding 1,000 MW

60. In the Bhopal Gas Disaster, the catalytic runaway exothermic reaction inside Tank 610 was triggered by the accidental entry of:

(A) Liquid nitrogen
(B) Water, reacting exothermically with methyl isocyanate in the presence of trace metallic contaminants
(C) Methane gas
(D) Concentrated sulfuric acid

61. During the thermal runaway in Tank 610 in Bhopal, temperatures exceeded 200 deg C, causing methyl isocyanate (MIC) to rapidly trimerize into:

(A) Toxic dimethyl isocyanate and phosgene gas
(B) Solid, non-toxic cyclic dimethylurea and methyl isocyanate trimer
(C) Hydrogen cyanide and methylamine vapors
(D) Methane and nitrogen gas

62. In seismology, the Moment Magnitude Scale (Mw) calculates earthquake magnitude based on the physical seismic moment M_0, defined as:

(A) M_0 = mu * A * D (where mu is rock shear modulus, A is rupture fault area, and D is average slip displacement)
(B) M_0 = Total weight of displaced soil
(C) M_0 = Acceleration of ground surface in m/s2
(D) M_0 = Depth of earthquake focus alone

63. According to the Bureau of Indian Standards (IS 1893:2016), India is divided into how many Seismic Zones?

(A) Two zones (Low and High)
(B) Four zones: Zone II (Low), Zone III (Moderate), Zone IV (Severe), and Zone V (Very Severe)
(C) Ten seismic zones
(D) Twenty micro-zones

64. Which Indian geographic regions fall under Seismic Zone V, representing the highest risk of catastrophic earthquake shaking?

(A) The entire Himalayan belt, the Northeast Indian states, and the Rann of Kutch (Gujarat)
(B) The Deccan plateau of Maharashtra and Karnataka
(C) The coastal beaches of Tamil Nadu
(D) The central plains of Madhya Pradesh

65. In slope engineering, the stability of a hillslope against landslides is evaluated by the Factor of Safety (FoS), defined as:

(A) Driving gravitational shear stress / Resisting shear strength
(B) Resisting shear strength of soil / Driving gravitational shear stress
(C) Slope height / Slope width
(D) Water infiltration rate / Evaporation rate

66. A hillside slope is on the verge of imminent catastrophic landslide failure when its Factor of Safety (FoS):

(A) Exceeds 2.5
(B) Equals or falls below 1.0
(C) Rises above 5.0
(D) Is exactly 1.5

67. The Sendai Framework for Disaster Risk Reduction (2015-2030) establishes four priority action areas, prioritizing:

(A) Understanding disaster risk, strengthening disaster governance, investing in resilience, and enhancing preparedness to 'Build Back Better'
(B) Total military mobilization during emergencies
(C) Building high concrete sea walls across all global coastlines
(D) Banning all human habitation within 100 km of mountains

68. In disaster management, the sequential cycle of phases is universally structured as:

(A) Mitigation -> Preparedness -> Response -> Recovery
(B) Destruction -> Evacuation -> Burial -> Abandonment
(C) Warning -> Alarm -> Defense -> Reconstruction
(D) Insurance -> Lawsuit -> Compensation -> Rebuilding

69. Which disaster phase focuses on long-term structural and non-structural measures designed to reduce the hazard impact before a disaster strikes?

(A) Disaster Response
(B) Disaster Mitigation
(C) Disaster Relief
(D) Disaster Search and Rescue

70. The Indian Disaster Management Act was enacted by Parliament in which year, establishing NDMA, SDMA, and DDMA?

(A) 1984
(B) 1999
(C) 2005
(D) 2015

71. The phenomenon where water-saturated, cohesionless sandy soil temporarily loses all shear strength during seismic shaking and behaves as a liquid is:

(A) Solifluction
(B) Soil Liquefaction
(C) Lateral spreading
(D) Slump failure

72. The destructive long-period ocean gravity waves generated by vertical tectonic displacement of the seabed during undersea earthquakes are:

(A) Tidal waves
(B) Tsunamis
(C) Wind waves
(D) Swell surges

73. In the deep open ocean, a tsunami wave travels at speeds comparable to a commercial jet airliner (700 to 900 km/h) with a wave height of:

(A) Only 0.5 to 1.0 meter (often undetectable by passing ships)
(B) Over 50 meters
(C) 100 meters
(D) Zero

74. As a tsunami wave enters shallow coastal waters, its speed decreases while its wave height grows dramatically due to the conservation of energy, termed:

(A) Shoaling effect
(B) Diffraction
(C) Refraction
(D) Dispersion

75. The National Institute of Disaster Management (NIDM), responsible for disaster human resource development and training in India, is located in:

(A) Mumbai
(B) New Delhi
(C) Hyderabad
(D) Pune

76. A chemical manufacturing facility releases 50 metric tons of methane (CH4, GWP100 = 28) and 2.0 metric tons of nitrous oxide (N2O, GWP100 = 273) annually. What is the total operational greenhouse gas emission footprint in metric tons of CO2 equivalent (t CO2e)?

(A) 1,400 t CO2e
(B) 1,946 t CO2e
(C) 2,200 t CO2e
(D) 2,850 t CO2e

77. Atmospheric CO2 concentration increases from pre-industrial baseline C_0 = 280 ppm to a modern level C = 420 ppm. Using the radiative forcing formula Delta F = 5.35 * ln(C / C_0) [given ln(1.50) = 0.4055], what is the net radiative forcing?

(A) 1.15 W/m2
(B) 2.17 W/m2
(C) 3.70 W/m2
(D) 4.52 W/m2

78. If the equilibrium climate sensitivity parameter lambda = 0.80 deg C / (W/m2), what is the projected equilibrium global mean surface temperature increase (Delta T) for a CO2 doubling radiative forcing Delta F = 3.7 W/m2?

(A) 1.50 deg C
(B) 2.20 deg C
(C) 2.96 deg C (approx 3.0 deg C)
(D) 4.50 deg C

79. Surface ocean water pH drops from pre-industrial 8.25 to present 8.14 (Delta pH = -0.11). Given 10^0.11 = 1.288, what is the exact percentage increase in hydronium ion concentration [H+] in surface seawater?

(A) 11.0%
(B) 15.5%
(C) 28.8%
(D) 45.0%

80. Compare two earthquakes of Richter magnitude M1 = 7.0 and M2 = 5.0 (Delta M = 2.0 units). Based on the Gutenberg-Richter energy formula log10(E) = 4.8 + 1.5 * M, how many times more seismic energy is released by the magnitude 7.0 earthquake?

(A) 10 times
(B) 100 times
(C) 1,000 times (10^3)
(D) 31.6 times

81. In a homogeneous granite rock mass (density rho = 2,700 kg/m3), Bulk Modulus K = 40 GPa and Shear Modulus G = 30 GPa. What is the compressional primary seismic wave velocity v_p = sqrt[(K + 4/3 * G) / rho]?

(A) 3.33 km/s
(B) 5.44 km/s
(C) 7.25 km/s
(D) 9.10 km/s

82. For the rock in Question 81, what is the shear secondary seismic wave velocity v_s = sqrt(G / rho)?

(A) 2.15 km/s
(B) 3.33 km/s
(C) 4.50 km/s
(D) 5.44 km/s

83. A seismograph station records the arrival of a P-wave (velocity = 6.0 km/s) followed by an S-wave (velocity = 3.5 km/s) with a time delay (t_s - t_p) of 20 seconds. What is the distance d to the earthquake epicenter [d = Delta t / (1/v_s - 1/v_p)]?

(A) 84 km
(B) 120 km
(C) 168 km
(D) 240 km

84. A dry, cohesionless sandy hillslope is inclined at slope angle beta = 30 degrees (tan 30 = 0.577). The internal friction angle of the sand is phi = 35 degrees (tan 35 = 0.700). What is the Factor of Safety (FoS = tan phi / tan beta)?

(A) 0.82
(B) 1.00
(C) 1.21
(D) 1.55

85. If heavy monsoon rains saturate the slope in Question 84 completely with groundwater rising to the ground surface, the effective normal stress is halved due to pore water buoyancy (gamma' / gamma_sat = 0.50). What is the new Factor of Safety?

(A) 0.61 (Imminent slope failure)
(B) 1.00
(C) 1.21
(D) 1.50

86. A 50 MW grid-connected wind energy farm generates 120,000 MWh (1.2 * 10^8 kWh) of clean electricity annually, displacing baseline coal electricity with an emission factor of 0.85 kg CO2/kWh. How many metric tons of CO2 offset credits (CERs) are generated each year?

(A) 51,000 CERs
(B) 102,000 CERs
(C) 150,000 CERs
(D) 204,000 CERs

87. Storage Tank 610 in the Bhopal plant contained 42 metric tons of liquid methyl isocyanate (density = 923 kg/m3). If the total physical tank volume was 57 m3, what was the liquid fill percentage (violating the safety operational ceiling of 50%)?

(A) 55.0%
(B) 65.5%
(C) 79.8%
(D) 95.0%

88. A natural lake of volume 1,000,000 m3 (10^9 Liters) has its water pH dropped from 6.5 to 4.5 by acid rain deposition. The net hydrogen ion excess is 3.13 * 10^-5 equivalents/Liter. How many kilograms of pure agricultural lime (CaCO3, equivalent weight = 50 g/eq) are required to neutralize the lake acidity?

(A) 500 kg
(B) 1,000 kg
(C) 1,565 kg (1.57 tons)
(D) 3,130 kg

89. A 700-meter-deep oceanic mixed layer undergoes a uniform thermal warming Delta T = 0.50 deg C. If the volumetric thermal expansion coefficient of seawater alpha = 2.0 * 10^-4 K^-1, calculate the resulting sea level rise Delta h = alpha * H * Delta T:

(A) 3.5 cm
(B) 7.0 cm (0.07 m)
(C) 14.0 cm
(D) 25.0 cm

90. In surface ocean water: [Ca2+] = 0.0103 mol/kg, and [CO3^2-] = 1.8 * 10^-4 mol/kg. If the apparent stoichiometric solubility product K'_sp for aragonite at 25 deg C is 6.6 * 10^-7, calculate the aragonite saturation state Omega = [Ca2+] * [CO3^2-] / K'_sp:

(A) 0.85
(B) 1.50
(C) 2.81
(D) 4.50

91. The energy release rate of a category 5 tropical cyclone with central pressure drop Delta P = 100 hPa and maximum winds of 65 m/s across a 300 km radius typically reaches:

(A) 10^6 Watts
(B) 10^9 Watts
(C) 10^13 Watts (exceeding human global electricity capacity)
(D) 10^20 Watts

92. To transition from global emissions of 51 Gt CO2e/year in 2020 down to Net-Zero emissions by 2050 (a 30-year period) through linear decarbonization, what is the required annual global reduction rate?

(A) 0.5 Gt CO2e/yr
(B) 1.7 Gt CO2e/yr
(C) 3.5 Gt CO2e/yr
(D) 5.1 Gt CO2e/yr

93. Trichlorofluoromethane (CFC-11) is the global reference compound with Ozone Depletion Potential ODP = 1.0. What is the ODP of Halon-1301 (CF3Br), reflecting the extreme catalytic destructive efficiency of atomic bromine?

(A) 0.5
(B) 1.0
(C) 10.0
(D) 100.0

94. Under Section 26 of the National Green Tribunal Act 2010, what is the maximum financial penalty the Tribunal can impose on a corporate enterprise for failing to comply with an environmental award or order?

(A) 10 lakh rupees
(B) 1 crore rupees
(C) 25 crore rupees (250 million INR)
(D) 100 crore rupees

95. According to the barometric inverse pressure effect in ocean dynamics, for every 1 hPa (1 millibar) drop in atmospheric pressure at the center of a cyclone, the ocean sea level rises by approximately:

(A) 1.0 mm
(B) 1.0 cm
(C) 10.0 cm
(D) 1.0 meter

96. A hydrological river flood with a 100-year return period (T = 100 years) has what annual exceedance probability in any single given year?

(A) 0.1%
(B) 1.0% (P = 1/T)
(C) 5.0%
(D) 10.0%

97. Over a standard 30-year home mortgage duration, what is the cumulative statistical probability [P = 1 - (1 - 1/T)^n] that a structure situated within a 100-year floodplain will experience at least one flood [given (0.99)^30 = 0.7397]?

(A) 1.0%
(B) 10.0%
(C) 26.0%
(D) 50.0%

98. Under the Indian EIA Notification 2006, all thermal power generation projects with capacities equal to or exceeding what threshold are classified as Category A projects requiring central MoEFCC clearance?

(A) 50 MW
(B) 100 MW
(C) 500 MW
(D) 1,000 MW

99. The Coastal Regulation Zone (CRZ) rules mandate that in CRZ-III (rural coastal zones), no new development construction is permitted within a 'No Development Zone' (NDZ) extending how many meters from the High Tide Line (HTL)?

(A) 10 meters
(B) 50 meters
(C) 200 meters
(D) 1,000 meters

100. The Sendai Framework for Disaster Risk Reduction establishes which specific target regarding disaster early warning systems by 2030?

(A) Substantially increase the availability of and access to multi-hazard early warning systems and disaster risk information to people
(B) Replace all early warning satellites with land sirens
(C) Discontinue international meteorological data sharing
(D) Limit warnings to capital cities alone

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Formulate the physical mechanisms governing the Enhanced Greenhouse Effect, detailing infrared absorption bands of greenhouse gases and Radiative Forcing.

Q2. Define Global Warming Potential (GWP) and Atmospheric Lifetime, and compare the radiative efficiencies of CO2, CH4, N2O, and SF6.

Q3. Analyze the chemical equilibria of Ocean Acidification, the decline in carbonate saturation state (Omega), and the physiological mechanisms of coral bleaching.

Q4. Formulate the Chapman Photochemical Mechanism of stratospheric ozone, and derive the catalytic ozone destruction cycles by halogen radicals.

Q5. Explain the atmospheric dynamics and heterogeneous chemistry of the Antarctic Ozone Hole, detailing Polar Stratospheric Clouds (PSCs).

Q6. Evaluate the Montreal Protocol (1987) and Kigali Amendment (2016), explaining Ozone Depletion Potential (ODP) and HFC phase-down.

Q7. Detail the institutional architecture of the Kyoto Protocol (1997), differentiating Annex I and Non-Annex I nations, and the three Flexible Market Mechanisms.

Q8. Analyze the legal and policy framework of the 2015 Paris Agreement, detailing Nationally Determined Contributions (NDCs) and Article 6 market mechanisms.

Q9. Compare the Representative Concentration Pathways (RCPs) and Shared Socioeconomic Pathways (SSPs) used in IPCC Assessment Reports.

Q10. Examine the constitutional environmental jurisprudence of India, analyzing Article 21, Article 48A, and Article 51A(g), and landmark Supreme Court rulings.

Q11. Analyze the statutory regulatory architecture of the Water (Prevention and Control of Pollution) Act 1974, detailing Consent to Establish (CTE) and SPCB powers.

Q12. Examine the provisions of the Air (Prevention and Control of Pollution) Act 1981, including Air Pollution Control Areas and emission standards.

Q13. Analyze the overarching mandate of the Environment (Protection) Act, 1986, Section 3 Central Government powers, and Section 5 binding directions.

Q14. Examine the regulatory mechanisms of the Forest (Conservation) Act, 1980, restrictions on forest diversion, and Compensatory Afforestation (CAMPA).

Q15. Detail the jurisdiction, powers, and procedure of the National Green Tribunal (NGT) under the NGT Act 2010, citing landmark environmental rulings.

Q16. Detail the four sequential stages of Environmental Impact Assessment (EIA) under the EIA Notification 2006: Screening, Scoping, Public Consultation, and Appraisal.

Q17. Explain the procedure, legal significance, and statutory exemptions of the Public Consultation process under the EIA Notification 2006.

Q18. Analyze the thermodynamic and meteorological physics of Tropical Cyclogenesis, and formulate the mechanics of Storm Surge amplification.

Q19. Detail Earthquake Seismology, distinguishing P-waves, S-waves, and Surface waves, and formulate the Moment Magnitude Scale (Mw).

Q20. Explain the Bureau of Indian Standards Seismic Zonation of India (IS 1893:2016), Zone Factors, and structural engineering mitigation.

Q21. Analyze the geotechnical physics of Soil Liquefaction during earthquakes, cyclic shear stresses, and mitigation via vibro-compaction.

Q22. Explain the hydrodynamics of Tsunamis, the conservation of energy in the Shoaling Effect, and DART ocean bottom early warning systems.

Q23. Formulate geotechnical slope stability, the Mohr-Coulomb failure criterion, Factor of Safety (FoS), and bioengineering slope stabilization.

Q24. Detail the chemical runaway kinetics of the Bhopal Gas Tragedy (1984), including water ingress, Tank 610 reactions, and safety system failures.

Q25. Analyze the long-term clinical toxicology, pulmonary pathophysiology, and genetic teratogenicity in survivors of the Bhopal Gas Disaster.

Q26. Detail the institutional architecture and statutory mandates under the Disaster Management Act, 2005: NDMA, SDMA, and DDMA.

Q27. Examine the Disaster Management Cycle, detailing the sequential activities across Mitigation, Preparedness, Emergency Response, and Recovery.

Q28. Analyze the four priority areas and seven global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030).

Q29. Differentiate the four typologies of Drought (Meteorological, Hydrological, Agricultural, Socioeconomic), and describe drought mitigation engineering.

Q30. Detail Quantitative Risk Assessment (QRA) for Major Accident Hazard (MAH) industrial chemical facilities: HAZOP and Fault Tree Analysis (FTA).

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[B]	**Q41:**	[A]	**Q61:**	[C]	**Q81:**	[B]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[B]	**Q62:**	[A]	**Q82:**	[B]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[B]	**Q63:**	[B]	**Q83:**	[C]
Q4:	[C]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[C]
Q5:	[A]	**Q25:**	[B]	**Q45:**	[C]	**Q65:**	[B]	**Q85:**	[A]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[B]	**Q66:**	[B]	**Q86:**	[B]
Q7:	[B]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[C]
Q8:	[C]	**Q28:**	[B]	**Q48:**	[B]	**Q68:**	[A]	**Q88:**	[C]
Q9:	[C]	**Q29:**	[B]	**Q49:**	[A]	**Q69:**	[B]	**Q89:**	[B]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[C]	**Q70:**	[C]	**Q90:**	[C]
Q11:	[A]	**Q31:**	[B]	**Q51:**	[A]	**Q71:**	[B]	**Q91:**	[C]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[A]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[B]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[C]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[C]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[B]	**Q95:**	[B]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[B]	**Q96:**	[B]
Q17:	[B]	**Q37:**	[A]	**Q57:**	[B]	**Q77:**	[B]	**Q97:**	[C]
Q18:	[A]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[C]	**Q98:**	[C]
Q19:	[B]	**Q39:**	[B]	**Q59:**	[B]	**Q79:**	[C]	**Q99:**	[C]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[C]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

- a_x: Radiative efficiency of gas x (W/m2 per kg of atmospheric concentration increase).
- C_x(t): Atmospheric mass of gas x remaining at time t following a pulse emission.
- a_{CO2} and C_{CO2}(t): Corresponding parameters for reference gas CO2.

2. Atmospheric Lifetimes and Removal Kinetics:

- Methane (CH4): Atmospheric lifetime ~11.8 years. Removed primarily by tropospheric gas-phase oxidation via hydroxyl radicals (.OH):

CH4 + .OH -> .CH3 + H2O.

- Nitrous Oxide (N2O): Atmospheric lifetime ~109 years. Unreactive in the troposphere; destroyed in the stratosphere by photolysis and reaction with excited oxygen atoms [O(.1D)].
- Sulfur Hexafluoride (SF6): Atmospheric lifetime ~3,200 years. Highly inert octahedron; destroyed only by high-energy photolysis in the mesosphere.
- Carbon Dioxide (CO2): Lacks a single simple first-order decay lifetime! It is governed by a sum of multi-timescale carbon cycle sinks (ocean mixed layer uptake: decades; deep oceanic circulation: centuries; mineral silicate weathering: tens of millennia).

3. Comparative Matrix (IPCC AR6 Metrics over 100-Year Horizon):

- Carbon Dioxide (CO2): Lifetime = Variable (multi-millennial); Radiative Efficiency = 1.37 * 10^-5 W/m2/ppb; GWP100 = 1.0 (Baseline).
- Methane (CH4): Lifetime = 11.8 years; Radiative Efficiency = 3.88 * 10^-4 W/m2/ppb; GWP100 = 27.9 (approx 28) (and GWP20 = 84 over 20-year horizon).
- Nitrous Oxide (N2O): Lifetime = 109 years; Radiative Efficiency = 3.00 * 10^-3 W/m2/ppb; GWP100 = 273.
- Sulfur Hexafluoride (SF6): Lifetime = 3,200 years; Radiative Efficiency = 0.57 W/m2/ppb; GWP100 = 23,500.

>>**Solution to Question 3:** Analyze the chemical equilibria of Ocean Acidification, the decline in carbonate saturation state (Omega), and the physiological mechan

calcification escalates, resulting in thinned shells and skeletal dissolution.
4. Molecular Mechanism of Thermal Coral Bleaching:
- Scleractinian corals maintain an obligate mutualistic endosymbiosis with photosynthetic dinoflagellates (zooxanthellae, Symbiodiniaceae) living within their gastrodermal cells.
- Elevated Sea Surface Temperature (SST > 1 to 2 deg C above summer maximum): Causes thermal damage to zooxanthellae Photosystem II (D1 protein denaturation).
- Photoinhibition triggers excessive production of toxic Reactive Oxygen Species (ROS, .O2-, H2O2, .OH) within algal chloroplasts, which leak into coral host cells.
- To prevent oxidative necrosis, the coral animal host expels its zooxanthellae symbionts. The coral tissue becomes transparent, revealing the white calcium carbonate skeleton underneath ('bleached'). Lacking algal photosynthetic glucose, corals starve and suffer mass mortality within weeks.

>>**Solution to Question 4:** Formulate the Chapman Photochemical Mechanism of stratospheric ozone, and derive the catalytic ozone destruction cycles by haloge

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>>Solution to Question 5: Explain the atmospheric dynamics and heterogeneous chemistry of the Antarctic Ozone Hole, detailing Polar Stratospheric Clouds (PS

Soviet bloc). Subject to legally binding emission caps (Assigned Amount Units, AAUs).
- Non-Annex I Parties: Developing nations (including India, China, Brazil). Exempted from binding emission caps during the first commitment period to prioritize economic poverty eradication.

3. The Three Flexible Market-Based Mechanisms:
Designed to lower the economic cost of achieving emission targets by allowing Annex I countries to achieve emission reductions extraterritorially where marginal abatement costs are lowest:
- 1. Clean Development Mechanism (CDM, Article 12):
* Operational Mechanism: An Annex I government or corporation invests in an emission-reduction or renewable energy project located in a Non-Annex I developing country (e.g., building a wind farm, solar park, or biomass plant in India).
* Certification: The project must satisfy the 'Additionality' criterion (proving emissions are lower than baseline business-as-usual).
* Output: The UN CDM Executive Board issues Certified Emission Reductions (CERs, where 1 CER = 1 metric ton CO_2e reduced). Annex I parties use CERs to offset domestic compliance targets.
- 2. Joint Implementation (JI, Article 6):
* An Annex I country finances an emission-reduction project in another Annex I country (primarily Eastern European EIT states).
* Output: Generates Emission Reduction Units (ERUs).
- 3. International Emissions Trading (IET, Article 17):
* 'Cap-and-Trade': An Annex I party that reduces its emissions below its statutory cap can sell its surplus Assigned Amount Units (AAUs) to another Annex I country that is struggling to meet its compliance targets.

>>Solution to Question 8: Analyze the legal and policy framework of the 2015 Paris Agreement, detailing Nationally Determined Contributions (NDCs) and Article

- Article 6.4 (Centralized Global Carbon Crediting Mechanism):
Replaces the Kyoto CDM with a centralized supervisory body that accredits projects and issues international carbon credit units verified against stringent baselines and sustainable development co-benefits.
- Article 6.8 (Non-Market Approaches): Facilitates international cooperation via finance, technological capacity building, and clean energy transfer without commercial carbon credit trading.

Constitutional Environmental Jurisprudence of India: Fundamental Rights, Duties, and Judicial Precedents:

1. Constitutional Foundations (The 42nd Constitutional Amendment Act, 1976):

The 42nd Amendment introduced explicit environmental conservation obligations into the Constitution of India:

- Article 48A (Directive Principles of State Policy):

'The State shall endeavour to protect and improve the environment and to safeguard the forests and wild life of the country.' Imposes an affirmative constitutional mandate upon legislative and executive branches.

- Article 51A(g) (Fundamental Duties):

'It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures.'

2. Judicial Expansion of Article 21 (The Right to Life):

Article 21 guarantees that 'No person shall be deprived of his life or personal liberty except according to procedure established by law.'

Through creative judicial activism, the Supreme Court of India broadened the meaning of 'Life' beyond mere animal existence to encompass the Right to a Wholesome, Clean, and Pollution-Free Environment:

- Subhash Kumar v. State of Bihar (1991): The Supreme Court formally ruled that the Right to Life under Article 21 includes the fundamental right to the enjoyment of pollution-free water and air for full enjoyment of life.

3. Integration of International Environmental Doctrines into Indian Law:

- The Precautionary Principle and Polluter Pays Principle:

In Vellore Citizens' Welfare Forum v. Union of India (1996), the Supreme Court held that the Precautionary Principle and the Polluter Pays Principle are essential features of Sustainable Development and are part of the environmental law of India under Article 21.

- The Public Trust Doctrine:

In M.C. Mehta v. Kamal Nath (1997), the Court imported the Roman law doctrine: Natural resources (rivers, forests, seashores) are held by the State in public trust for the collective benefit of the entire public, and the State cannot convert public trusts into private commercial ownership.

- Absolute Liability Doctrine:

In M.C. Mehta v. Union of India (Oleum Gas Leak Case, 1987), Justice P.N. Bhagwati formulated the principle of Absolute Liability: An enterprise engaged in a hazardous or inherently dangerous industry owes an absolute and non-delegable duty to the community, subject to zero exceptions (unlike Rylands v. Fletcher strict liability).

>>**Solution to Question 11:** Analyze the statutory regulatory architecture of the Water (Prevention and Control of Pollution) Act 1974, detailing Consent to Establish

works (ETPs/STPs), monitors sewage and trade effluent discharges, and enforces compliance with environmental standards.

3. The Dual 'Consent' Regulatory Regime:

Under Section 25, no person or corporate enterprise can establish or operate any industrial plant, operation, or disposal system likely to discharge sewage or trade effluent without prior statutory consent from the SPCB:

- 1. Consent to Establish (CTE / NOC):

* Obtained BEFORE initiating site ground construction, structural engineering, or machinery installation.
* Evaluates site suitability, proximity to water bodies, and the engineering adequacy of proposed effluent treatment plants (ETPs).

- 2. Consent to Operate (CTO):

* Obtained AFTER constructing ETP facilities but BEFORE commissioning commercial production.
* Prescribes quantitative effluent discharge standards (pH, TSS, BOD, COD, heavy metals) and requires installation of Continuous Online Effluent Monitoring Systems (OCEMS). CTO must be renewed periodically.

4. Coercive Enforcement Powers:

- Power of Entry and Inspection (Section 23): Authorized officers have unrestricted right to enter any industrial facility to inspect systems and take wastewater samples (Section 21) following strict legal sampling protocols admissible in judicial courts.
- Power to Issue Binding Directions (Section 33A):

Empowers the SPCB to issue binding written directions to any person, officer, or authority, including:

* The immediate closure, prohibition, or regulation of any industrial operation or process.
* The stoppage or regulation of electricity, water, or other municipal utility services to defaulting units.

- Criminal Penalties (Sections 41 to 44): Operating without consent or violating discharge standards carries mandatory imprisonment of 1.5 to 6 years and substantial monetary fines.

>>Solution to Question 32: Examine the provisions of the Air (Prevention and Control of Pollution) Act 1981, including Air Pollution Control Areas and emission sta

(cyclones, baghouse filters, scrubbers, ESPs), installation of sampling ports and platforms on chimney stacks, and adherence to National Ambient Air Quality Standards (NAAQS) and point-source emission limits.

4. The 1987 Statutory Amendment: Inclusion of Noise Pollution:

- Section 2(a) was formally amended in 1987 to expand the legal definition of 'air pollutant' to include: 'any solid, liquid or gaseous substance including noise present in the atmosphere in such concentration as may be or tend to be injurious to human beings or other living creatures or plants or property.'
- Enabled the enactment of the Noise Pollution (Regulation and Control) Rules, 2000, establishing ambient day and night decibel standards across Industrial, Commercial, Residential, and Silence Zones.

>> Solution to Question 13: Analyze the overarching mandate of the Environment (Protection) Act, 1986, Section 3 Central Government powers, and Section 5 bind

- E-Waste Management Rules 2022.
- Plastic Waste Management Rules 2016.

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1. Statutory Genesis and International Standing:
Established on October 18, 2010, under the National Green Tribunal Act 2010 pursuant to Article 21 constitutional jurisprudence, India became only the third nation in the world (after Australia and New Zealand) to establish a specialized environmental tribunal.
2. Specialized Composition:
- Unique Multi-Disciplinary Bench: Every NGT bench comprises an equal balance of:
* Judicial Members: Retired judges of the Supreme Court or Chief Justices of High Courts (headed by the Chairperson in Principal Bench, New Delhi).
* Expert Members: Highly qualified scientists and technical experts holding doctorates in environmental engineering, ecology, or forestry with over 15 years of practical field experience.
- Eliminates judicial dependence on outside expert witnesses, ensuring fast, scientifically robust adjudication.
3. Statutory Jurisdiction (Section 14, 15, and 16):
- Original Jurisdiction (Section 14): Over all civil cases where a substantial question relating to the environment (including environmental damage and public health) is involved, arising out of the implementation of seven specified environmental statutes:
* 1. Water Act, 1974
* 2. Water Cess Act, 1977
* 3. Forest (Conservation) Act, 1980
* 4. Air Act, 1981
* 5. Environment (Protection) Act, 1986
* 6. Public Liability Insurance Act, 1991
* 7. Biological Diversity Act, 2002
(Note: Wildlife Protection Act 1972 and Indian Forest Act 1927 are explicitly excluded from NGT jurisdiction).
- Relief and Compensation (Section 15): Directing restitution of damaged ecosystems, relief to pollution victims, and clean-up costs.
- Appellate Jurisdiction (Section 16): Hearing statutory appeals challenging Environmental Clearances (EC) or CTO rejections within 30 days.
4. Landmark NGT Precedents:
- Ban on 10-Year-Old Diesel Vehicles in Delhi-NCR (Vardhaman Kaushik case): Prohibited older diesel vehicles to curb toxic PM2.5 air pollution.
- Ban on Rat-Hole Coal Mining in Meghalaya: Halted hazardous unscientific coal mining causing severe acid mine drainage that turned the Kopili and Lukha rivers toxic.
- Art of Living Yamuna Floodplain Fine: Imposed environmental restoration restitution fees of 5 crore rupees for damaging sensitive riparian floodplains during the World Culture Festival.

>>**Solution to Question 16:** The four sequential stages of Environmental Impact Assessment (EIA) under the EIA Notification 2006: Screening, Scoping, Publ

Impact Assessment Authority (SEIAA) upon recommendation of the State Expert Appraisal Committee (SEAC).

3. The Four Sequential Stages of Environmental Clearance:

- Stage 1: Screening (Applicable to Category B Projects Only):

* The SEAC scrutinizes the preliminary Form 1 application to determine whether the proposed project requires full EIA studies (categorized as Category B1) or does not require an EIA study (categorized as Category B2, which proceeds directly to appraisal).
* Category A projects bypass screening; a full comprehensive EIA is mandatory.

- Stage 2: Scoping:

* The Expert Appraisal Committee determines detailed, comprehensive Terms of Reference (ToR) addressing all potential environmental impacts (air dispersion modeling, hydrogeological surveys, biodiversity assessment, socio-economic baseline, risk analysis).
* The project proponent conducts seasonal baseline field monitoring (excluding monsoon) to draft the comprehensive Environmental Impact Assessment (EIA) report and Environmental Management Plan (EMP).

- Stage 3: Public Consultation:

* Comprises two components: A formal public hearing conducted at the physical project site, and inviting written responses from stakeholders to ensure public participation and transparency.

- Stage 4: Appraisal:

* Detailed multidisciplinary scrutiny by the EAC/SEAC of the final EIA report, outcome of the public consultations, and the commitments made by the project proponent.
* The EAC/SEAC makes formal recommendations to the regulatory authority (MoEFCC/SEIAA) to grant Environmental Clearance with specific environmental safeguards, or reject the application.

>>Solution to Question 17 Explain the procedure, legal significance, and statutory exemptions of the Public Consultation process under the EIA Notification 2006.

4. Statutory Exemptions from Public Hearing:
Under Paragraph 7(i) III of the EIA Notification 2006, the public hearing component is explicitly exempted for:
- Projects located within officially notified industrial estates or export processing zones where the estate itself has undergone prior environmental clearance.
- All Category B2 projects.
- Highway expansion projects that do not involve land acquisition.
- Modernization or expansion of irrigation projects.
- Defense, national security, or strategic projects as determined by the Central Government.

>>Solution to Question 18: Analyze the thermodynamic and meteorological physics of Tropical Cyclogenesis, and formulate the mechanics of Storm Surge amplific

by 5 to 10 meters, flooding low-lying coastal plains.

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of 40 to 50 mm/year creates immense crustal stress accumulation along the Himalayan Frontal Thrust (HFT), Main Boundary Thrust (MBT), and Main Central Thrust (MCT), making the Indian subcontinent highly seismically vulnerable.

2. The Four Seismic Zones of India (IS 1893:2016 Part 1):

India is categorized into four seismic zones, with corresponding Zone Factors (Z, representing peak ground acceleration PGA as a fraction of gravity):

- Zone II (Low Seismic Hazard, Z = 0.10):

* Geographic Coverage: Stable central and southern Deccan Plateau (parts of Karnataka, Andhra Pradesh, Tamil Nadu, Madhya Pradesh).

* Expected Intensity: MSK VI or less.

- Zone III (Moderate Seismic Hazard, Z = 0.16):

* Geographic Coverage: Western Maharashtra (including Mumbai, Koyna), Kerala, Goa, parts of Gujarat, Punjab, and Uttar Pradesh.

* Expected Intensity: MSK VII.

- Zone IV (Severe Seismic Hazard, Z = 0.24):

* Geographic Coverage: Northern Indo-Gangetic Plains, Delhi-NCR, parts of Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Bihar, Sikkim, and the Rann of Kutch.

* Expected Intensity: MSK VIII.

- Zone V (Very Severe / Catastrophic Seismic Hazard, Z = 0.36):

* Geographic Coverage: Entire Northeastern Indian states (Assam, Meghalaya, Arunachal, Nagaland, Manipur, Mizoram, Tripura), parts of Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Andaman & Nicobar Islands, and Bhuj (Kutch, Gujarat).

* Expected Intensity: MSK IX and above.

3. Earthquake-Resistant Structural Engineering Principles:

- Ductility and Confinement: Enforcing IS 13920 ductile detailing in reinforced concrete (RC) frames. Providing close stirrup spacing (shear ties) in beam-column junctions to prevent brittle shear failure.

- Base Isolation Systems: Employs elastomeric lead-rubber bearings (LRBs) beneath building foundations to uncouple superstructure from horizontal ground vibrations.

- Symmetrical Structural Configuration: Avoiding soft storeys (open ground floor parking) and structural torsional irregularities that amplify earthquake collapse.

>>**Solution to Question 21:** Analyze the geotechnical physics of Soil Liquefaction during earthquakes, cyclic shear stresses, and mitigation via vibro-compaction.

* Under rapid cyclic seismic shaking, cyclic loading occurs too fast for pore water to drain away (undrained conditions).
* Pore water is compressed, causing excess pore water pressure (Delta u) to escalate rapidly.
* When excess pore water pressure equals total overburden stress (u = sigma_total): Effective stress becomes zero: sigma' = 0!
* Because effective inter-particle contact stress is zero, the soil shear strength collapses to zero: tau_f = 0.

3. Surface Manifestations and Structural Failures:
- Sand Boils and Mud Volcanoes: High subterranean hydrostatic pressure forces liquefying sand slurries upward through ground fissures, erupting onto the surface.
- Loss of Bearing Capacity: High-rise buildings tilt, rotate, or sink bodily into the liquefied ground without structural fracture (e.g., Niigata earthquake, 1964).
- Lateral Spreading: Gentle coastal slopes liquefy and slide horizontally toward rivers or seas, shearing bridge piers and underground utility pipelines.

4. Ground Improvement Mitigation Engineering:
- Vibro-Compaction and Stone Columns: Inserting large vibrating poker probes into loose sand deposits, combined with water/air jets and gravel backfill. Vibrations densify sand relative density from Dr < 40% to Dr > 80%, eliminating contractive volume collapse.
- Geotechnical Drainage: Installing vertical earthquake drains (prefabricated perforated geosynthetic wick drains) that dissipate excess pore water pressure instantaneously during shaking, preventing u from reaching total stress.

>>Solution to Question 22 Explain the hydrodynamics of Tsunamis, the conservation of energy in the Shoaling Effect, and DART ocean bottom early warning syst

* The front of the wave slows down while the rear wave continues at high speed, compressing wave energy horizontally.
* The wave height (amplitude H) grows from 0.5 m into a catastrophic vertical wall of water reaching 10 to 30 meters height, violently inundating coastal settlements.

4. DART (Deep-ocean Assessment and Reporting of Tsunamis) Detection Networks:

- Bottom Pressure Recorders (BPRs): Highly sensitive piezoelectric quartz crystal pressure gauges anchored to the deep seabed (up to 6,000 m depth). Can detect tsunami water column pressure variations as small as 1 millimeter!
- Acoustic Uplink & Satellite Link: The BPR transmits acoustic telemetry signals to a surface moored buoy, which immediately relays real-time tsunami height and arrival time data via NOAA/Iridium satellites to national disaster warning centers within minutes.

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1. Industrial Chemical Profile: Methyl Isocyanate (MIC):
Methyl Isocyanate (CH3-N=C=O, MW = 57.05 g/mol) was synthesized at the Union Carbide India Limited (UCIL) Bhopal facility as a chemical intermediate in the manufacture of the carbamate pesticide Carbaryl (Sevin):
- Synthesis: Methylamine reacts with Phosgene:
CH3NH2 + COCl2 -> CH3NH-CO-Cl + HCl -> CH3-N=C=O (MIC) + 2 HCl
- Physical Properties: Highly volatile liquid (boiling point 39.1 deg C, vapor pressure 348 mmHg at 20 deg C). Extremely reactive, toxic, and lachrymatory.
2. The Fatal Chemical Runaway in Storage Tank 610:
On the night of December 2, 1984, approximately 500 to 1,000 kg of water accidentally entered underground stainless steel Storage Tank 610 during a routine filter line water-washing operation (due to missing slip-blind isolation plates):
- Exothermic Water Hydrolysis:
CH3-N=C=O + H2O -> [CH3-NH-COOH] (Unstable methylcarbamic acid)
[CH3-NH-COOH] -> CH3NH2 (Methylamine) + CO2 (g) + Heat (-84 kJ/mol).
- Autocatalytic Secondary Reactions:
The generated methylamine reacted violently with remaining MIC:
CH3NCO + CH3NH2 -> CH3-NH-CO-NH-CH3 (1,3-Dimethylurea) + Heat.
- Catalytic Trimerization:
Corrosion in the tank wall had released iron rust contaminants (Fe3+). In the presence of ferric catalysts and elevated temperatures, MIC underwent explosive trimerization:
3 CH3NCO --[Fe3+, Heat]--> (CH3NCO)3 (Trimethyl isocyanurate) + Heat (-134 kJ/mol).
- Thermal Runaway: Because heat generation far exceeded the tank's dissipation capacity, internal temperature surged above 200 deg C, and internal pressure skyrocketed to 55 psi (3.8 bar), violently rupturing the tank safety relief valve.
3. Catastrophic Multiple Safety System Failures:
- 1. Tank Refrigeration System: Designed to keep MIC stored at 0 deg C to suppress runaway reactions; had been decommissioned months earlier to save electricity costs!
- 2. Vent Gas Scrubber: Designed to neutralize venting gases with 10% caustic soda (NaOH); was unpowered and out of service.
- 3. Flare Tower: Designed to combust leaking toxic gases; disconnected for maintenance.
- 4. Water Curtain Sprayers: Water pressure was insufficient to reach the elevated gas discharge stack!
- Consequence: Over 40 metric tons of lethal MIC and breakdown products (HCN, phosgene) vented into the cool night air, drifting over densely populated informal settlements, causing immediate death of over 3,800 people and chronic disability in over 500,000 survivors.

>>**Solution to Question 25:** Analyze the long-term clinical toxicology, pulmonary pathophysiology, and genetic teratogenicity in survivors of the Bhopal Gas Disaste

2. Systemic Cyanide Toxicity Hypothesis:
Thermal decomposition of MIC inside Tank 610 produced hydrogen cyanide (HCN). Post-mortem examinations by toxicologists revealed cherry-red venous blood and elevated rhodanese enzyme activity, and administration of sodium thiosulfate ($Na_2S_2O_3$, an antidote facilitating cyanide conversion to thiocyanate) produced clinical improvements in early survivors.
3. Chronic Long-Term Clinical Pathologies:
- Respiratory: Severe persistent Pulmonary Fibrosis, Chronic Obstructive Pulmonary Disease (COPD), restrictive lung disease, and chronic bronchiolitis obliterans.
- Immunological: Chronic immune suppression, elevated circulating autoantibodies, and destruction of cellular immunity.
- Neurological: Memory loss, sleep disturbances, peripheral neuropathy, and severe chronic depression.
4. Genetic and Teratogenic Damage:
- High rates of spontaneous miscarriages (spontaneous abortion rate exceeded 43% in pregnant women exposed during the tragedy).
- Chromosomal Aberrations: Cytogenetic studies revealed chromosomal breaks, sister chromatid exchanges (SCE), and translocations in peripheral blood lymphocytes that persisted for decades.
- Congenital Deformities: Elevated incidence of congenital neural tube defects, musculoskeletal deformities, and congenital heart disease in children born to exposed mothers.

>>Solution to Question 26: ...the institutional architecture and statutory mandates under the Disaster Management Act, 2005: NDMA, SDMA, and DDMA.

- Dedicated Disaster Funds: Statutory creation of the National Disaster Response Fund (NDRF) and National Disaster Mitigation Fund (NDMF) mirrored at State and District tiers.

>>Solution Question 25: Analyze the Disaster Management Cycle and its alignment with the Sendai Framework for Disaster Risk Reduction (2015-2030) Response, a

of persons, businesses, communities and countries.'
2. The Four Priority Action Areas:
- Priority 1: Understanding Disaster Risk:
Promoting the collection, analysis, and dissemination of disaggregated disaster data; mapping multi-hazard geospatial vulnerabilities and climate risk modeling.
- Priority 2: Strengthening Disaster Risk Governance to Manage Disaster Risk:
Enacting robust legislative frameworks, empowering decentralized local authorities, and fostering multi-sectoral public-private coordination.
- Priority 3: Investing in Disaster Risk Reduction for Resilience:
Allocating dedicated public capital and mobilizing private investment into structural resilience, retrofitting critical infrastructure (schools, hospitals), and nature-based solutions.
- Priority 4: Enhancing Disaster Preparedness for Effective Response and to 'Build Back Better' in Recovery, Rehabilitation, and Reconstruction:
Equipping emergency responders, conducting international mutual-aid training, and utilizing post-disaster recovery to upgrade infrastructure resilience.
3. The Seven Global Targets (A to G):
- Target A: Substantially reduce global disaster mortality by 2030.
- Target B: Substantially reduce the number of affected people globally by 2030.
- Target C: Reduce direct disaster economic loss in relation to global Gross Domestic Product (GDP).
- Target D: Substantially reduce disaster damage to critical infrastructure and disruption of basic services (health and educational facilities).
- Target E: Substantially increase the number of countries with national and local disaster risk reduction strategies.
- Target F: Substantially enhance international cooperation to developing countries.
- Target G: Substantially increase the availability of and access to multi-hazard early warning systems (MHEWS) and disaster risk information.

>>**Solution to Question 29:** Differentiate the four typologies of Drought (Meteorological, Hydrological, Agricultural, Socioeconomic), and describe drought mitigation

economic goods (food grains, drinking water, hydroelectric power), resulting in food price inflation, agrarian distress, farmer suicides, and mass rural distress migration.

3. Engineering and Watershed Mitigation Strategies:

- Micro-Irrigation Systems: Drip and sprinkler irrigation achieve water application efficiencies exceeding 90% (compared to 30-40% for flood irrigation), delivering water directly to root zones.
- Integrated Watershed Development: Constructing check dams, continuous contour trenches (CCTs), percolation ponds, and gabion structures across micro-catchments to intercept monsoon runoff and recharge groundwater aquifers.
- Drought-Tolerant Crops: Promoting climate-resilient coarse grains and millets (Sorghum, Pearl Millet, Finger Millet) with high water-use efficiency.

>>Solution to Question 30: Quantitative Risk Assessment (QRA) for Major Accident Hazard (MAH) industrial chemical facilities: HAZOP and Fault Tree Anal

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